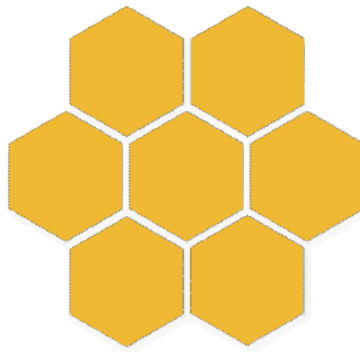


Google Summer of Code 2025

(The HoneyNet Project)

Glutton : Implementing Protocol Parsers for Glutton Using Spicy



Personal Info

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Nationality : India

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University Info

University: Indian Institute of Technology (IIT), Roorkee

Major: Production and Industrial Engineering

Current Year: 2nd year (May 2027 expected graduation)

Degree: Bachelor of Technology (B.Tech)

1) Top project choice :

Glutton continues to be my preferred framework for this project. The plan is to integrate the Spicy Parser into Glutton, enhancing the existing HTTP and DNS parsers.

The Glutton project description can be found [here](#).

2) Are you willing and able to work on other projects instead?

Yes, I would love to work on **Honeynet's IntelOwl Project**. Since I already have experience contributing to Honeynet, transitioning to this project will be smooth. The goal is to implement additional analyzers based on the existing ones.

The IntelOwl Project description can be found [here](#).

3) Please describe your preferred coding languages and experience.

I have hands-on experience working with **Go** on various projects and have previously contributed to **Glutton**, which has given me a solid understanding of its code structure and implementation. This familiarity allows me to work with and contribute to the project efficiently. Additionally, I possess a strong background in the **MERN stack**, which enables me to build full-stack applications, and proficiency in **C++**, which provides a solid foundation for **systems-level programming**.

Beyond this, I have also delved into **blockchain technology**, gaining practical experience in its development. I'm particularly interested in exploring **blockchain security**, as it's an area with immense potential in the evolving tech landscape. I actively participate in **Capture the Flag (CTF)** competitions to further expand my knowledge. These challenges allow me to sharpen my skills in **networking, cybersecurity, and various other technical areas**, helping me stay up to date with the latest trends and tools in the field of security.

4) Please describe any Windows, Unix, or Mac OS X development experience relevant to your chosen project. If your project does not require OS-specific expertise, feel free to leave it empty.

This particular project demands a solid understanding and experience in **Linux networking**. In the past, I've worked on a **web server in Go**, where I built a [Library Management Server](#), which provided me with valuable hands-on experience in server-side development and networking concepts. Additionally, I have contributed to Glutton, and to better grasp its underlying principles, I studied "**Beej's Guide to Networking**" which significantly deepened my understanding of networking protocols and concepts.

Furthermore, I have built various **small servers** and **packet sniffers** in Go as part of personal projects, which allowed me to explore networking at a deeper level and gain practical experience. These projects helped me to understand better how network protocols work and how to handle networking tasks efficiently. Along with my experience, I've developed a strong ability to troubleshoot network issues and optimize performance. My continuous learning in Linux networking and hands-on project work have equipped me with the skills necessary to excel in this project.

5) Please describe any previous usage with Honeynet Project tools or honeypots in general

I have previous experience working with Honeynet Project tools, particularly with **Glutton**. Glutton is a powerful tool designed to emulate services and capture malicious interactions, allowing for effective monitoring and analysis of cyberattacks. My work with Glutton involved deploying it in various environments to gather intelligence on attacker behavior and studying different attack patterns.

Additionally, I have worked with another honeypot tool - **Cowrie**, which provided valuable insights into common attack vectors, such as SMB brute force attempts and malware delivery mechanisms.

6) Please describe any previous Honeynet Project or honeypot related development experience, including details of any patches, code or ideas you may have previously submitted.

I have been contributing to and analysing the code of Glutton since **January 2025**. Interacting with [@glaslos](#) and [@fursiyya](#)., I was able to understand and solve the following issues:

- **Implementing the Parser for iSCSI protocol** : [Here](#) (Merged)
- **Drop privileges if started as root** : [Here](#) (Open)

7) Please describe any previous open source development experience, including projects you have worked on.

I have worked on the following projects :

- **TollChain** : It was our submission for EthIndia 2024. [TollChain](#) is a blockchain-based reimagining of FASTag that combines RFID technology with blockchain security, which ensures privacy through anonymous authentication while maintaining transparency through the blockchain's immutable ledger.
- **Ch3ckm8** : Written in Go, [ch3ckm8](#) is a chess engine in Go using bitboards for efficient move generation and alpha-beta pruning for optimized move search—integrated in-game visualization of FEN strings, utilizing piece square tables for dynamic position evaluation.
- **Quizio** : Quizio is an online quizzing platform that lets you host quizzes, allow proctoring during the quiz, and review the answers. (Backend : [Athena](#) , Frontend : [Helios](#))

8) What school do you attend and what is your specialty/major at the school?

I am a **second-year** Production and Industrial Engineering student at the **Indian Institute of Technology, Roorkee**.

9) How many years have you attended there?

I have attended two years with two more to go.

10) What city/country will you be spending this summer in?

I will be residing in [Roorkee, Uttarakhand \(University\), India](#), or [New Delhi, Delhi \(Home\), India](#) during the summer, depending on the summer break dates of my university.

11) How much time do you expect to have for this project?

As a college student, I intend to work on the project during my summer breaks from **May to July**. After completing my University exams in April, I will start working in May. I can dedicate **40 hours a week from May to July**, while in **August**, after the classes commence, I can dedicate about **25 hours a week**.

12) Please list all jobs, summer classes, vacations and/or other commitments that you'll need to work around.

There are **no exams or planned vacations** throughout the coding period. I generally work from **4 PM to 2 AM IST (6:30 AM to 4:30 PM Eastern Time)**, although I find it flexible to adjust my working hours if required.

Besides this project, I have **no commitments/vacations planned for the summer**. I shall keep my status posted to all the community members every week and **maintain transparency** in the project.

13) Have you participated in any previous Summer of Code projects? If so please describe your projects and experience, including what you liked or didn't like about the experience.

No, I have not participated in any previous Summer of Code Projects. I am looking forward to a wonderful experience with Honeynet and Glutton.

14) Have you applied for (or intend to apply for) any other Google Summer of Code 2025 projects? If so, which ones?

I have not applied for any other Google Summer of Code Projects. **I am 100% dedicated to Glutton and have no plans to submit a proposal to any other organisation or project.**

15) If you have a URL for your resume/CV, please list it here.

My resume can be found [here](#).

16) Please describe your proposed project in detail, including deliverables and expected timeline with milestones (this is the long answer, so spend most time here!).

The proposed project aims to implement a Go wrapper around spicy-based protocol parsers for HTTP and DNS.

Spicy is designed for protocol analysis, parsing, and handling complex data formats. It provides high-level constructs for working with binary data, network protocols, and protocol-specified state machines. It supports bit-level parsing. Since Spicy is more focused on the declarative syntax than the mechanics of parsing the data, we can focus more on the logic and structure of the parsed data.

Now, according to [RFC 2616](#), the structure of an HTTP request is as follows:

```
Request      = Request-Line           ; Section 5.1
               *( ( general-header    ; Section 4.5
                   | request-header    ; Section 5.3
                   | entity-header ) CRLF) ; Section 7.1
               CRLF
               [ message-body ]       ; Section 4.3
```

To parse this request, the idea is to mirror the working of the **http.ReadRequest()** and **net.SplitHostPort()** in the **net/http** and **net** packages respectively. We will start with the Parser architecture design.

We will define the following units : **(Customized this [example](#))**

```

module MyHTTP;

const Token      = /^[^ \t\r\n]+/;
const WhiteSpace = /[ \t]+/;
const NewLine    = /\r?\n/;

type Version = unit {
  :      /HTTP\//;
  number: /[0-9]+\.[0-9]+/;
};

type URI = unit {
  path: /^[^? ]+;/
  query: optional /(?!<=\\?).+;/
};

type RequestLine = unit {
  method: Token;
  :      WhiteSpace;
  uri:    URI;
  :      WhiteSpace;
  version: Version;
  :      NewLine;

  on %done {
    print self.method, self.uri, self.version.number;
  }
};

public type Request = unit {
  request_line: RequestLine;
  headers: Headers;
  body: bytes &length=self.content_length;

  var content_length: uint64 = 0; // Need it to parse the exact length of body correctly
};

```

The following spicy grammar outlines the basic structure of a HTTP request. The entry point of the parser would be at the **Request unit** which would further structure the request as defined.

Each header will be parsed in name-value pairs.

```

type Headers = unit {
  headers: Header[] &until($$.name == b"");

  on %done {
    for (h in self.headers) {
      if ( to_lower(h.name) == b"content-length" ) {
        self.outer.content_length = atoi(h.value);
      }
    }
  }
};

type Header = unit {
  name: /^[^:\r\n]+/;
  :    /:[ \t]*/;
  value: /^[^r\n]*/;
  :    /\r\n/;
};

```

Once the request is appropriately parsed, we can move on to **POST body analysis** of some exceptional cases like Ethereum RPC calls or the Hadoop Yarn hack etc.

```
if ( self.request_line.method == b"POST" ) {
    # Ethereum RPC detection
    if ( self.body.contains(b"eth_blockNumber") ) {
        print "Ethereum RPC call detected";
    }

    # Hadoop YARN detection
    if ( self.request_line.uri.path.contains(b"/cluster/apps/new-application") ) {
        print "Hadoop YARN request detected";
    }
}
```

We can also call the proper handler function for these two cases (already defined in Glutton).

The succeeding step would be to generate appropriate responses to the received requests for example :

```
procedure trigger_smb_handler() {
    local smb_config = b'''
    HTTP/1.1 200 OK
    Server: Apache
    Content-Type: text/plain; charset=UTF-8
    Content-Length: 93

    [global]
    encrypt passwords = yes
    name resolve order = lmhosts wins host bcst
    ''';
    self.connection.send(smb_config);
}
```

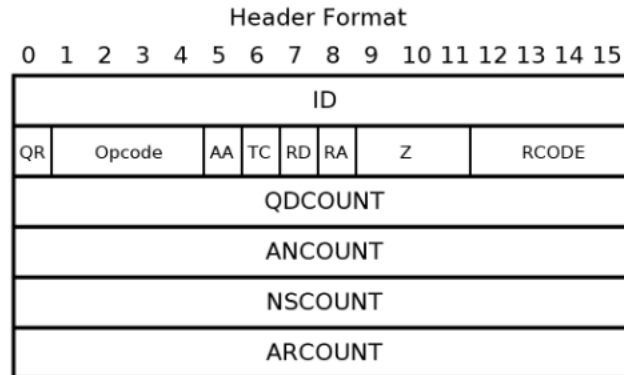
Procedures for each type of response can be created this way. An event-based approach can be used to integrate the existing handlers in Glutton. (See below)

```
if (self.request_uri.starts_with(b"/vpn/")) {
    emit citrix_smb_request(self.connection);
}

event citrix_smb_request(conn: connection);
```

This event can now be handled in the Golang part.

Now, similarly, for DNS, the request header structure is as follows



It can be parsed as the following unit

```
type DNS_Header = unit {  
  id: uint16;  
  flags: uint16 &convert=DNS_Flags($$);  
  qdcount: uint16;  
  ancourt: uint16;  
  nscount: uint16;  
  arcourt: uint16;  
};
```

The whole DNS request can now be similarly structured.

```
public type DNS_Packet = unit {  
  header: DNS_Header;  
  questions: DNS_Question[header.qdcount];  
  answers: DNS_Record[header.ancourt];  
  authorities: DNS_Record[header.nscourt];  
  additionals: DNS_Record[header.arcourt];  
};
```

Once the spicy grammar is programmed, we can create an executable using [spicyc -j](#). This can then be used to integrate with Glutton.

For testing purposes, we can either have unit tests set up for each parser like the existing ones, or we can compare the result from both the parsers (The existing one in Golang and the Spicy Parser)

```
goOutput, goErr := parseWithGo(payload)
spicyOutput, spicyErr := parseWithSpicy(payload)
if (goErr != nil) != (spicyErr != nil) {
    t.Errorf("Error mismatch: Go: %v , Spicy: %v", goErr, spicyErr)
}
if !compareOutputs(goOutput, spicyOutput) {
    t.Errorf("Output mismatch: Go: %+v, Spicy: %+v", goOutput, spicyOutput)
}
```

DELIVERABLES

- 1) **HTTP Parser** : Complying with RFC 2616 and already implemented exceptional cases in Glutton.
- 2) **DNS Parser** : Supporting all the RR types.
- 3) **Testing Methodology** : Unit Tests and Differential Tests (Comparing outputs from Spicy Parser to Go Parsers)
- 4) **Documentation** : Documenting the process of creating a new parser and its integration with Glutton.

TIMELINE

GSOC TIMELINE	
Pre-GSOC Period	
April 8 - May 1	Read about HTTP/DNS Parsers along with practicing Spicy Grammar
Community Bonding Period	
May 1 - May 26	- Get to know more about the Glutton community. - Discuss with the mentor to finalize the architecture of the project.

Coding Period	
May 27 - June 2	- Get familiar with the architecture of Glutton and its implementation. - Explore the implementation of the existing parsers and the Go Libraries.
Phase 1 : HTTP Parser	
June 3 - June 15	- Create a basic HTTP parser in compliance with RFC 2616 and the existing parser.
June 16 - June 20	- Focus on handling special cases like Ethereum rpc calls and yarn attacks.
June 21 - June 25	- Create a basic script for response generation. - Write an event based approach with hooks to utilise some of the existing code.
June 26 - June 30	- Write tests for the parser
Phase 2 : DNS Parser	
July 1 - July 13	- Create a basic DNS parser in compliance with RFC 1035.
Mid-Term Evaluation	
July 14 - July 21	- Handle compression and EDNS
July 21 - July 25	- Create a script to detect attacks using hooks and generate appropriate responses
July 26 - July 31	- Write tests for the parser
Phase 3 : Integration	
Aug 1 - Aug 7	- Integrate with existing Glutton code.
Aug 8 - Aug 20	- Performance Optimization
Aug 21 - Aug 25	- Write Documentation
Final Evaluation	

FUTURE GOALS

- 1) **Developer Toolkit for New Parsers** : Create a developer-friendly toolkit to streamline the creation of new Spicy parsers for additional protocols
- 2) **Expand Protocol Coverage** : Extend Glutton's capabilities by implementing Spicy parsers for additional protocols commonly targeted in attacks
- 3) **Integrate CI/CD Pipeline** : Automate the build, test, and deployment process for Glutton and its Spicy parsers by integrating a CI/CD pipeline using Github Actions

17) Why are you well suited to perform this project and why are you interested in it?

With **more than a year** of experience in developing full-stack applications with Go, Javascript, Typescript, C/C++, and various other tech stacks, both as personal projects and contributing to large open-source software, I'm always eager to broaden my knowledge by diving into new technologies. My expertise can meet the high standards required in open-source organizations. I can work effectively within a team, provide and receive constructive feedback, and adapt to changing project requirements.

As an **active member of the Glutton community**, I'm familiar with your work culture, codebase structure, and how to navigate the community for support. My approach towards the project showcases my understanding of the project's goals and my ability to take the initiative.

Experience building production-level applications at **SDSLabs** has equipped me to handle technical challenges and adapt to evolving project requirements. My strong problem-solving skills, honed through academic challenges, ensure I can tackle complex tasks effectively.

I'm a highly motivated individual with a passion for learning new technologies. Given the opportunity to contribute to this Honeynet project, I'm dedicated to making it exceptionally user-friendly and maximizing its impact.

18) Have any of our members met you face to face, such as at one of our recent public events? If so, please list who/where.

I haven't met them, but I have had insightful discussions with both **Lukas, the maintainer of Glutton, and Muhammad Bilal Arif, the project mentor**. These interactions have been valuable in shaping my understanding of Glutton and refining my approach to the project.

